MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 9 (18-22 Oct 2026) - Period 1	Ch 8 - 8.6 Selection: IF statements

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 9 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 85% DIV/MOD correct; ^ vs ** errors in code. Today: relational operators and IF...THEN...ELSE...ENDIF.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use relational operators (= < <= > >= <>) and write IF...THEN...ELSE...ENDIF in pseudocode and Python if/elif/else to make decisions. Syllabus 8.1.4(b), 8.1.4(f); AO2, AO3.

Success criteria — *what will successful learning look / sound like?*

I can write a correct IF statement with ELSE in Cambridge pseudocode; I can code it in Python with correct indentation; I test both branches.

Key knowledge / skill / understanding:

Selection chooses a path based on a Boolean condition; <> means not equal (!= in Python); indentation defines blocks in Python.

Likely misconception / access or challenge need:

= compares in Python (it assigns); ELSE needs a condition; only one IF can be used.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Storytelling **Time:** 6 min

What students will do:

A short story: a student arrives at the school gate - IF before 7:15 THEN 'on time' ELSE 'late slip'. Students turn the story into a flowchart decision.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Story slide; flowchart symbols.

Support (low performers / SEN):

Flowchart with decision box drawn.

Stretch (fast learners / gifted):

Add a third outcome (very late).

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Worked example then practice: grade calculator (mark -> A*, A, B, C, U) in pseudocode with nested IF/ELSE and Python if/elif/else. Test each boundary mark.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) 8.6; Python; boundary test table.

Support (low performers / SEN):

Template with first two branches written.

Stretch (fast learners / gifted):

Validate input is 0-100 before grading.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Think-pair-share **Time:** 8 min

What students will do:

Spot the error: 4 short IF snippets with faults (missing ENDIF, = vs ==, wrong boundary, bad indentation). Pairs fix and explain.

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Error-spotting sheet.

Support (low performers / SEN):

Errors highlighted, students explain fix.

Stretch (fast learners / gifted):

Write one buggy snippet for another pair.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 7 min

Hook: traffic light decision. Model relational operators, then IF...THEN...ELSE...ENDIF with 2-space indentation as in the syllabus, and Python if/elif/else. Think aloud on testing boundaries (e.g. 69, 70, 71).

What is new or innovative in this lesson? *(ETEC looks for this)*

Boundary testing taught explicitly with test tables, mirroring how Paper 2 marks careful condition writing.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Wrong boundary (> vs >=); missing colon or indentation; using = in conditions; order of elif tests wrong so A* never reached. Likely: 'my program always says U'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: write pseudocode IF for 'age >= 18 OUTPUT Adult ELSE Minor'; give 3 boundary test values. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.6 IF questions; program: input temperature, output 'Hot', 'Warm' or 'Cold'.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Gate story was instantly relatable; grade calculator engaged all. Error-spotting pairs mixing language groups worked well. 80% tested all boundaries.

What will I change next time?

Show the order-of-elif problem with a live failing example earlier. Provide boundary tables on paper.

Students needing follow-up:

10BR1, 10BR3: about 3 students per class still mix = and ==; quick retrieval starter next lesson.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H
Create	**Discussion**	**Pair / group**	**Learner choice**	**Self-assessment**	**Peer assessment**	**SEN / low**	**Gifted**
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 9 (18-22 Oct 2026) - Period 2	Ch 8 - 8.6 Selection: CASE statements

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 9 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 80% wrote correct IF; boundary choices weak in 10BR3. Today: CASE...OF...OTHERWISE...ENDCASE and when to choose CASE over IF.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students write CASE OF ... OTHERWISE ... ENDCASE in pseudocode, implement it in Python (match-case or if/elif), and justify CASE vs IF. Syllabus 8.1.4(b); AO2, AO3.

Success criteria — *what will successful learning look / sound like?*

I can write a CASE statement with an OTHERWISE clause; I can code it in Python; I can explain when CASE is clearer than nested IF.

Key knowledge / skill / understanding:

CASE selects one branch from many values of one variable; OTHERWISE must be last; Python 3.10+ match-case or if/elif.

Likely misconception / access or challenge need:

CASE can test any condition like IF; OTHERWISE is optional so never needed.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Role play **Time:** 7 min

What students will do:

'Customer service menu' role play: one student is the menu (1=Balance, 2=Top-up, 3=Help, other='Invalid'), others call in; class records the branches.

Evidence tool: Observation checklist

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Menu card; role cards.

Support (low performers / SEN):

Role card with script.

Stretch (fast learners / gifted):

Add a hidden option and discuss OTHERWISE.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Rewrite a nested-IF 'day of week' program as CASE in pseudocode, then code a canteen menu (CASE on item code) in Python with match-case or if/elif.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) 8.6 CASE; syllabus pseudocode guide p.44; Python 3.10+.

Support (low performers / SEN):

Side-by-side IF vs CASE template.

Stretch (fast learners / gifted):

Combine CASE with a loop later: menu repeats until 'Q'.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Discussion / dialogu **Time:** 7 min

What students will do:

Pairs decide: IF or CASE? for 5 scenarios (grade boundaries, menu choice, password check, month number, shipping zones) and justify.

Evidence tool: Exit ticket

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Scenario cards.

Support (low performers / SEN):

Scenarios with picture hints.

Stretch (fast learners / gifted):

Explain why ranges suit IF better.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 7 min

Retrieval: = vs == quick check. Hook: a phone menu 'press 1, press 2'. Model CASE OF Choice ... OTHERWISE OUTPUT "Invalid" ENDCASE (syllabus layout), then Python match-case and the if/elif equivalent.

What is new or innovative in this lesson? *(ETEC looks for this)*

Role-play models program flow physically before coding; students judge the best control structure, building AO3 evaluation.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

OTHERWISE not last; using ranges in CASE; Python version without match-case (fallback to if/elif); missing default. Likely: 'CASE is just shorter IF'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: write a 4-branch CASE in pseudocode; one sentence on CASE vs IF. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.6 CASE questions; program: input month number 1-12, output the season (handle invalid).

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Role play made OTHERWISE obvious. Students debated Hijri vs Gregorian months for the homework - good cultural link.

What will I change next time?

Check Python version in the lab (2 PCs below 3.10, no match-case); teach if/elif fallback first.

Students needing follow-up:

Students who missed the IF lesson (approx. 2 per class) need the worked example sheet.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☑C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☑3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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